

Top 10 open source deep learning tools

1. TensorFlow

TensorFlow (www.tensorflow.org) is an open source deep learning framework for the development of machine learning and deep learning models, founded on November 9, 2015.

Features

- Provides an end-to-end model, from building to deployment
- Supports model deployment on mobile devices and embedded devices also
- Good documentation and help from the community
- Extensive support for multiple GPUs
- Graph visualisation and queues
- Supports image processing, computer vision, natural language processing, text classification, speech recognition, etc

2. Keras

Keras (keras.io) is an open source deep learning library in Python, released on March 27, 2015.

Features

- Extensive documentation and developer guides
- Easy to learn and use
- Offers clear and comprehensive error messages
- Supports models for mobile devices, the Web as well as for Java Virtual Machine
- Enables distributed deep learning models on graphics processing units (GPUs) and tensor processing units (TPUs)
- Broadly adopted in the research community as well as in the industries

3. PyTorch

PyTorch (pytorch.org) is an open source machine learning library



Figure 1: Diverse deep learning applications

released on September 2016 and authored by Adam Paszke, Sam Gross, Soumith Chintala, and Gregory Chanan. The essential elements (cheat sheet) for new users are available at pytorch.org/tutorials/beginner/ptcheat.html.

Features

- Broadly accepted across major cloud platforms to provide easy development and straightforward scalability
- Facilitates end-to-end pipeline flow, from Python development

to mobile device deployment (iOS and Android)

- Active research community and industry developers provide robust ecosystem libraries that enable development in various domains — from computer vision to reinforcement learning
- Provides direct interface with ONNX (open neural network exchange) compatible systems, environments, visualisers, and more, by exporting models in the standard ONNX format
- Offers multiple GPU support for the implementation of deep learning models
- Supports C++ interface for high performance and low latency applications
- Supports data sharing with external libraries

4. OpenNN

OpenNN (www.opennn.net) is an open source neural networks library for machine learning and deep learning. Its first version (0.1) was released on November 22, 2018.

Latest applications that use deep learning

- Autonomous vehicles: Tesla and Waymo
- Polybee: Pollinating bee drone in greenhouses
- RNNoise: RNN based noise suppression library
- Grammarly: NLP based text summarisation
- Perlmutter: An Nvidia project that targets 3D mapping of the universe
- MERIDIAN: Helping compile a catalogue of all known fish sounds and creating a benchmark data set for marine bioacoustics
- Neurala Brain: Neural network and software that emulate the functions of the human brain for navigation and perception
- Zeroeyes: Real-time fire arms detection, prevention (police) and protection
- WaveNet: Generates raw audio from speech patterns using CNNs
- 6sense: A B2B sales and marketing platform that predicts buying behaviour
- Robbie AI: Patient monitoring platform to prevent falls and delirium in ICU patients